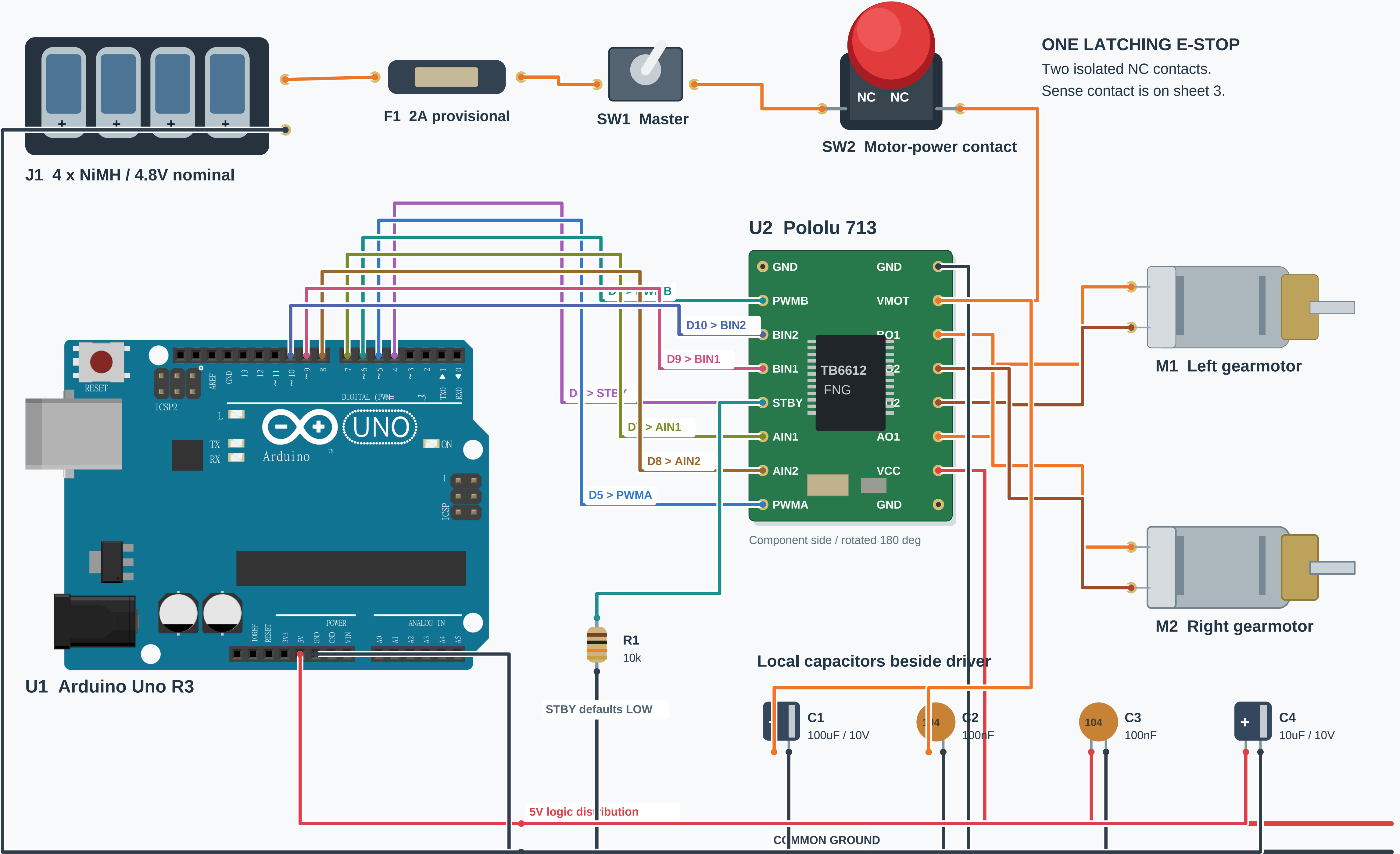


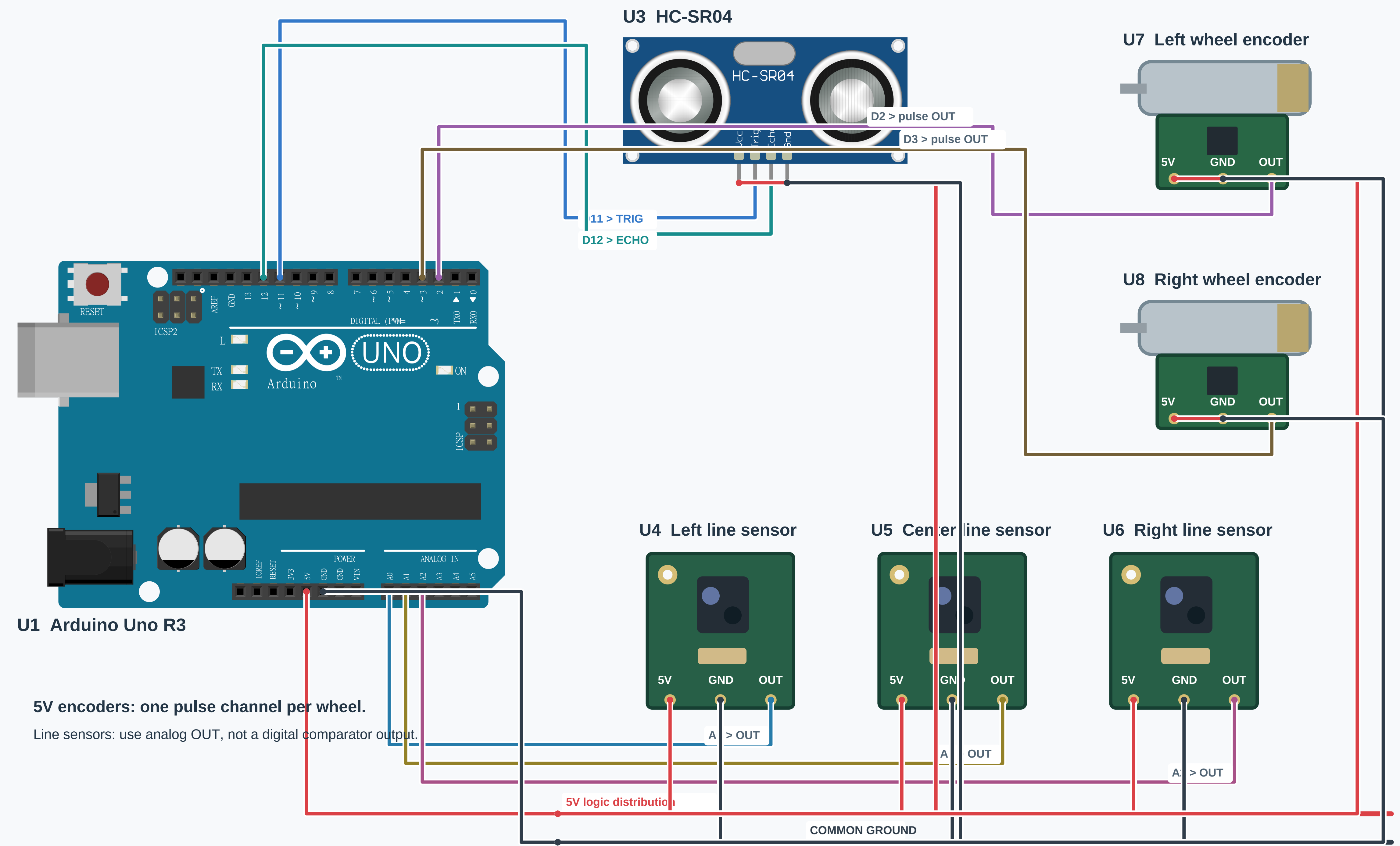
Motor drive and power cutoff

Recognizable components with wire-by-wire motor connections. The same Uno continues on the next two sheets.



Range, line sensors and wheel encoders

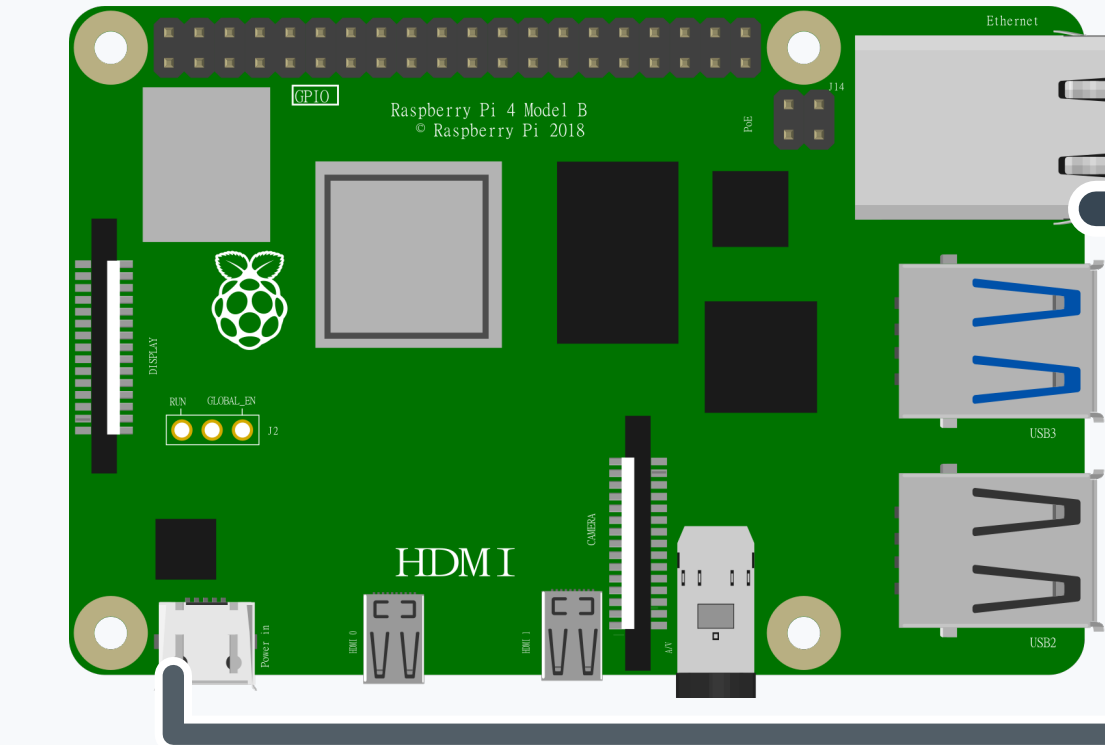
Signal wires terminate at the actual Uno and HC-SR04 connector locations. Power rails are wiring junctions, not a second supply.



Pi host, IMU and safety inputs

USB supplies the Uno and carries serial data. The IMU uses a 3.3V-compatible breakout and translated I2C signals.

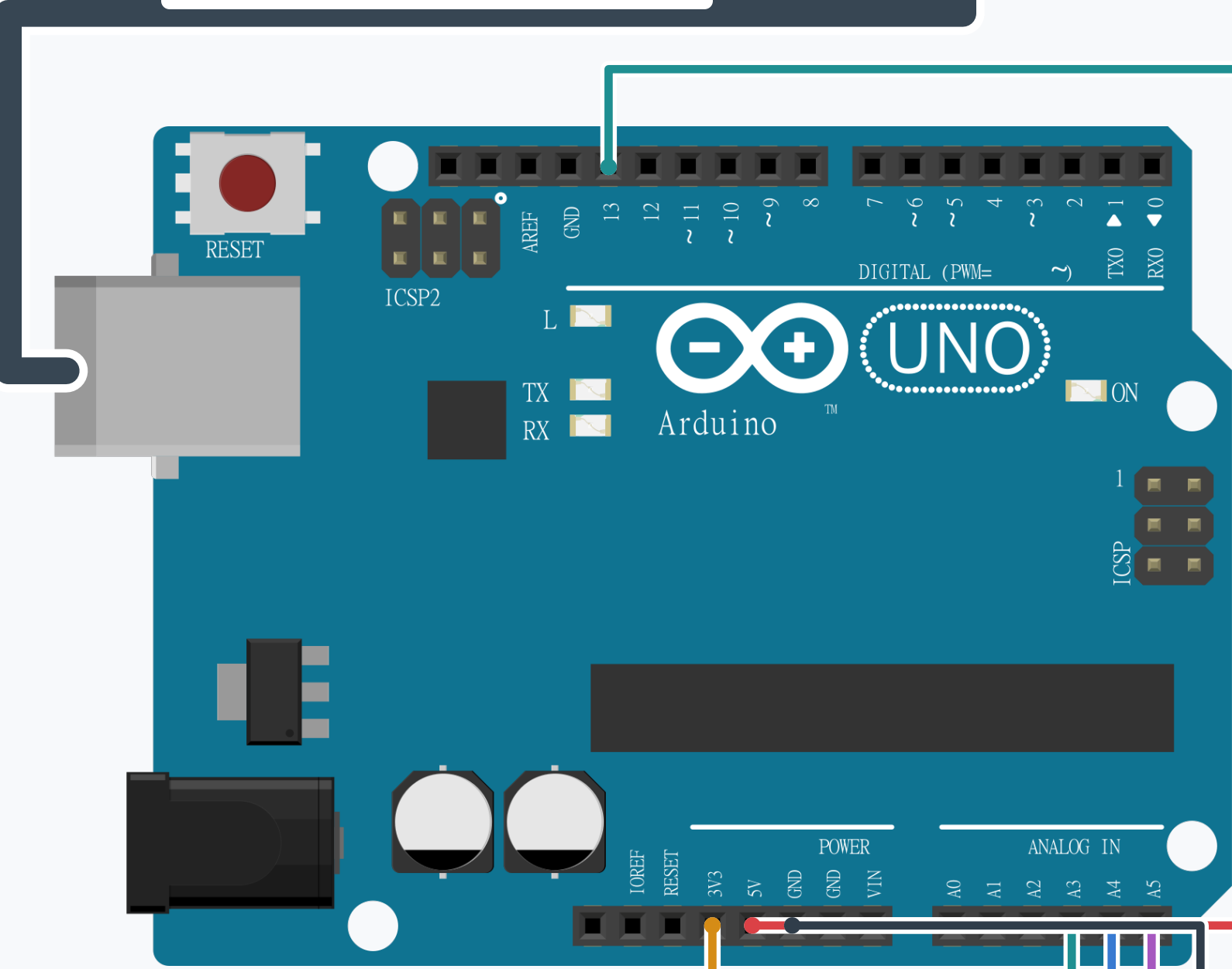
U9 Raspberry Pi 4B



5.1V / 3A
USB-C supply

Pi GPIO is unused.
Wi-Fi handles V2V.

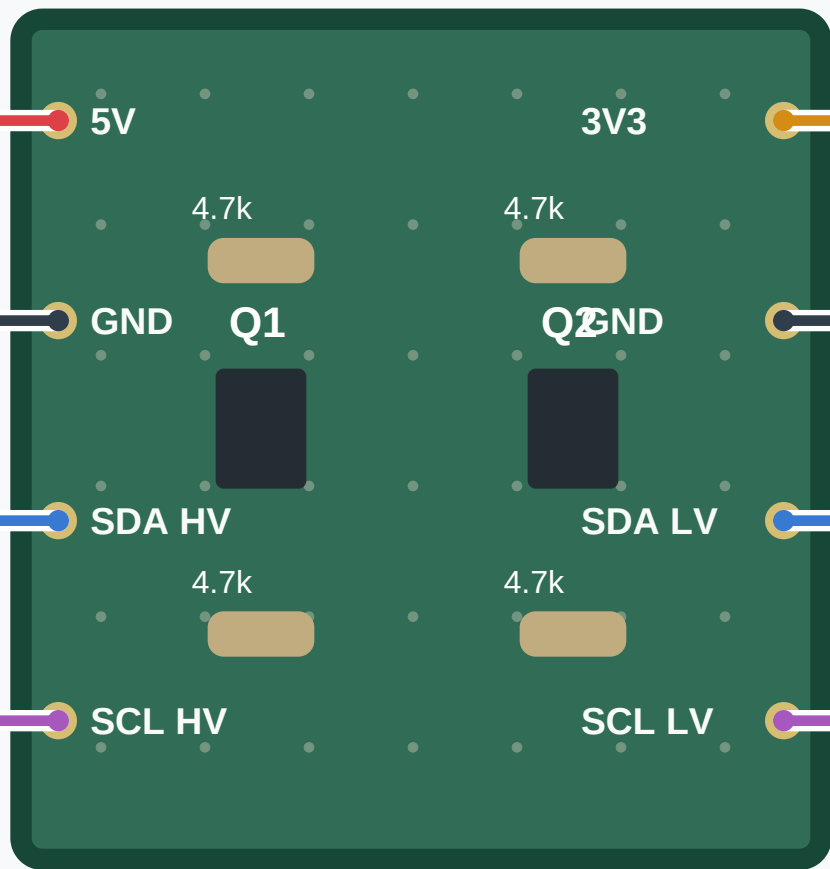
USB-A to USB-B: power + serial



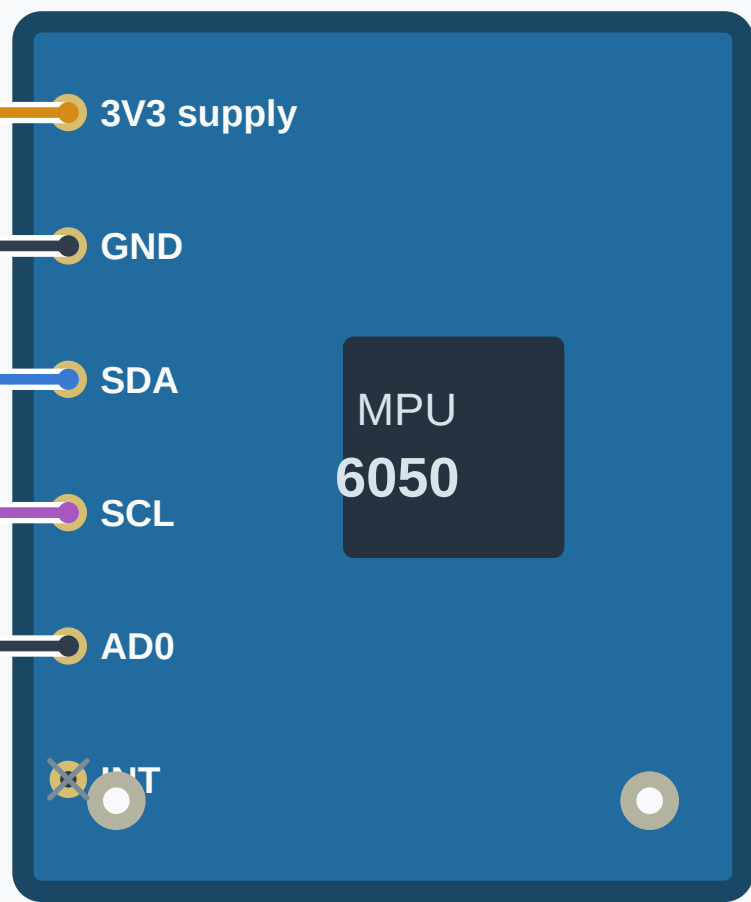
U1 Arduino Uno R3

3.3V supply

Q1/Q2 + R3-R5 I2C translator



U10 MPU6050 breakout



AD0 low = address 0x68

Same latching actuator as SW2 on sheet 1.



SW4 Fog demo

R2
10k



SW3 Sense contact

A4 > SDA
A5 > SCL

A3 > normally closed contact

5V logic distribution

COMMON GROUND